

Geometric Mean

geometric mean (also known as the **mean proportional**) is a mean or average which indicates a central tendency of a finite collection of positive real numbers by using the product of their values (as opposed to the arithmetic mean, which uses their sum).

$$\text{Geometric Mean: } \sqrt[3]{3 \times 9 \times 27} = \sqrt[3]{729} = 9 \quad \text{Arithmetic Mean: } (3 + 9 + 27) / 3 = 21$$

Statistical measure used to calculate the central tendency used to find the central value of the data set in statistics especially when dealing with **growth rates or percentages**. This method is great for comparing **values that change over time**, like investment returns or population growth.